

An Exact Stability, Crossing, and Jordan Atlas for Erlang-2 Distributed-Delay Feedback

Route-A source-local certificate HCS-C343

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Abstract

We solve a scalar negative-feedback equation with a normalized Erlang shape-two memory kernel. Compatible histories admit an exact three-dimensional linear-chain realization, whose cubic Routh determinant gives a necessary-and-sufficient stability threshold and a transverse imaginary crossing. We further exhaust every repeated-root surface, determine all Jordan sizes, count unstable roots, and give closed semigroup formulas on simple, double, and triple-root faces. Zero damping and feedback, the triple-root point, the excluded zero-rate kernel, the fast-memory limit, source ownership, collision boundaries, and the rejected Route-A decision are closed explicitly. No nonlinear periodic branch is claimed.

1 Frozen memory owner

Fix

$$a, b \geq 0, \quad r > 0, \quad K_r(s) = r^2 s e^{-rs} \quad (s \geq 0),$$

and consider

$$\dot{x}(t) = -ax(t) - b \int_0^\infty K_r(s)x(t-s) ds. \quad (1)$$

The kernel is normalized; it has mean $2/r$, variance $2/r^2$, and Laplace transform $r^2/(\lambda + r)^2$. A compatible fading-memory history is one for which the following integrals exist and permit the standard differentiation under the integral:

$$z_1(t) = \int_0^\infty r e^{-rs} x(t-s) ds, \quad z_2(t) = \int_0^\infty K_r(s)x(t-s) ds.$$

Lemma 1 (Exact linear chain). *For compatible initialization, equation (1) is equivalent to*

$$\dot{x} = -ax - bz_2, \quad \dot{z}_1 = r(x - z_1), \quad \dot{z}_2 = r(z_1 - z_2). \quad (2)$$

It is not asserted that arbitrary values of (z_1, z_2) encode a prescribed prehistory.

Proof. Integration by parts gives the two filter equations in (2); the first equation is (1). Conversely, subtract the displayed convolutions from a solution of the two stable filter equations. The differences solve homogeneous first-order equations and vanish initially, so they vanish for all later times. Thus compatible chain solutions recover the delay equation exactly. $\square$

Put $X = (x, z_1, z_2)^\top$. Then $\dot{X} = MX$, where

$$M = \begin{pmatrix} -a & 0 & -b \\ r & -r & 0 \\ 0 & r & -r \end{pmatrix},$$

and direct expansion gives

$$\begin{aligned} p(\lambda) &= \det(\lambda I - M) = (\lambda + a)(\lambda + r)^2 + br^2 \\ &= \lambda^3 + c_2\lambda^2 + c_1\lambda + c_0, \\ c_2 &= a + 2r, \quad c_1 = r(r + 2a), \quad c_0 = r^2(a + b). \end{aligned} \quad (3)$$

Theorem 2 (Sharp stability and imaginary crossing). *For $b > 0$, the zero solution is exponentially stable if and only if*

$$0 < b < b_H, \quad b_H = \frac{2(a+r)^2}{r}. \quad (4)$$

At $b = b_H$,

$$p(\lambda) = (\lambda + a + 2r)(\lambda^2 + r(r + 2a)), \quad (5)$$

so the roots are $-a - 2r$ and $\lambda_\pm = \pm i\omega_H$, with $\omega_H = \sqrt{r(r + 2a)}$. The conjugate pair is simple and crosses from left to right:

$$\operatorname{Re} \lambda'_+(b_H) = \frac{r^2}{2\{r(r + 2a) + (a + 2r)^2\}} > 0. \quad (6)$$

For $b > b_H$, exactly two roots lie in the open right half-plane.

Proof. For $b > 0$, all c_j are positive. The remaining cubic Routh–Hurwitz minor is

$$c_2 c_1 - c_0 = r\{2(a+r)^2 - br\}.$$

This proves the equivalence in (4). At equality, multiplication verifies (5). Implicit differentiation of $p(\lambda, b) = 0$ at $\lambda = i\omega_H$ yields (6). Above the threshold the first Routh column has signs $+, +, -, +$, hence two sign changes and exactly two open-right-half-plane roots. $\square$

The phrase *chain equivalence owner* records the first manuscript round: this is an all-parameter analytic theorem, not stability inferred from sampled roots.

2 Complete degeneracy and propagator atlas

Theorem 3 (All repeated roots and Jordan blocks). *In the domain $a, b \geq 0$, $r > 0$, all repeated-root cases are precisely:*

1. $b = 0$. The root $-r$ has one size-two Jordan block when $a \neq r$, and at $a = r$ the root $-r$ has one size-three block.
2. $0 \leq a < r$ and

$$b = b_D = \frac{4(r-a)^3}{27r^2} > 0. \quad (7)$$

Here

$$p(\lambda) = (\lambda - \mu)^2(\lambda - \nu), \quad \mu = -\frac{r+2a}{3}, \quad \nu = -\frac{4r-a}{3},$$

and the Jordan sizes are two and one.

The positive- b defective face lies strictly inside the stable region.

Proof. Differentiating (3) and taking the discriminant gives

$$p'(\lambda) = (\lambda + r)(3\lambda + r + 2a), \quad \text{disc } p = br^2\{4(r-a)^3 - 27br^2\}. \quad (8)$$

The first critical root belongs to p exactly when $b = 0$. The second belongs to p exactly on (7); it has positive b precisely when $a < r$. These observations exhaust (8), and substitution gives the factorization. Moreover,

$$\det[e_1, Me_1, M^2 e_1] = r^3 \neq 0.$$

Thus M is cyclic and its minimal polynomial equals p : each repeated root has one Jordan block of its algebraic multiplicity. Finally

$$\frac{b_D}{b_H} = \frac{2(r-a)^3}{27r(a+r)^2} < 1,$$

so the positive defective face is stable. $\square$

For distinct roots $\lambda_1, \lambda_2, \lambda_3$, Lagrange interpolation modulo p gives

$$e^{tM} = \sum_{j=1}^3 e^{t\lambda_j} \prod_{k \neq j} \frac{M - \lambda_k I}{\lambda_j - \lambda_k}. \quad (9)$$

On a double face, write $p = (\lambda - \mu)^2(\lambda - \nu)$ and set

$$P_\nu = \frac{(M - \mu I)^2}{(\nu - \mu)^2}, \quad P_\mu = I - P_\nu, \quad N_\mu = (M - \mu I)P_\mu.$$

Hermite interpolation yields

$$e^{tM} = e^{\mu t}(P_\mu + tN_\mu) + e^{\nu t}P_\nu, \quad N_\mu^2 = 0. \quad (10)$$

At $a = r, b = 0$, $N = M + rI$ satisfies $N^3 = 0$ and

$$e^{tM} = e^{-rt}(I + tN + t^2 N^2/2). \quad (11)$$

Equations (9)–(11) close the flow on every spectral stratum.

The phrase *Routh Hopf atlas owner* marks the substantive first revision: transversality, unstable-root count, discriminant exhaustion, Jordan sizes, and defective semigroups have all been added.

3 Boundaries and claim discipline

At $b = 0$, $p = (\lambda + a)(\lambda + r)^2$. Hence $a > 0$ is exponentially stable despite the defective $-r$ block. At $a = b = 0$, the simple zero root is a constant x -mode and the memory transient contains te^{-rt} . For $a = 0, b > 0$, Theorem 2 specializes to stability for $0 < b < 2r$, an imaginary pair at $b = 2r$, and two right-half-plane roots above it. The point $a = r, b = 0$ is the triple block in (11).

The value $r = 0$ is excluded: the displayed expression ceases to be a normalized Erlang density. For fixed bounded λ , $r^2/(\lambda + r)^2 \rightarrow 1$ as $r \rightarrow \infty$, so the bounded slow root tends to $-(a + b)$ while two memory modes are fast. This limiting observation is not substituted for any finite- r theorem.

Theorem 2 proves a simple transverse imaginary *linear spectral crossing*. A nonlinear vector field, its nondegeneracy coefficient, and its center-manifold reduction have not been specified. Therefore neither existence nor stability of a nonlinear periodic branch is claimed.

4 Evidence, sources, and collisions

The certificate carries six exact kernel rows, 120 rational Routh faces, 20 Hopf-factorization rows, and 30 repeated-root rows, totaling 2,017 audited scalar leaves. The independent checker, independent SymPy lane, two-directory byte replay, and repaired-hash hostile suite are executable receipts. Sampling does not prove the continuum theorem; the displayed Routh, discriminant, and cyclicity arguments do.

The Erlang linear-chain lineage is documented by Hurtado and Kiro Singh [1] and MacDonald [3]; Boese supplies primary gamma-delay stability context [2]. These records support provenance, not a priority claim for our formulas or assembly.

The nearest workspace owners are C210, a scalar *discrete* retarded delay with an infinite Lambert- W root ladder; C218, a Kelvin-Voigt wave PDE; and C272, an Erlang age-transport renewal PDE. None owns this normalized distributed-memory scalar feedback, its finite Markov realization, and complete cubic Hopf/Jordan atlas. This is a workspace collision statement only.

5 Route-A decision

The phrase *Jordan boundary and route firewall* marks the final revision. Under NO_BAD_EULER_OR_ROOT_NUMBER, the tuple is

$$(A0_FAIL, A1_FAIL, A2_FAIL, A3_FAIL, A4_FAIL).$$

There is no arithmetic origin or prime-power clock; linear modes are not an arithmetic primitive-orbit ledger; no dynamical Euler product, target determinant, functional equation, Weil compression, natural target-zero quantization, target-zero match, or Hilbert-Polya operator is supplied. Route A is rejected and Route B remains locked. No target arithmetic local data, Euler factor, root number, automorphy, divisor, or functional equation is asserted.

AI use. A generative language model assisted proof organization, code scaffolding, and manuscript drafting. The analytic derivation, independent checker, symbolic lane, hostile tests, and deterministic artifacts define the audit record.

References

- [1] P. J. Hurtado and A. S. Kiro Singh, *Generalizations of the Linear Chain Trick: incorporating more flexible dwell time distributions into mean field ODE models*, *Journal of Mathematical Biology* 79 (2019), 1831–1883. DOI: [10.1007/s00285-019-01412-w](https://doi.org/10.1007/s00285-019-01412-w).
- [2] F. G. Boese, *The stability chart for the linearized Cushing equation with a discrete delay and with gamma-distributed delays*, *Journal of Mathematical Analysis and Applications* 140 (1989), 510–536. DOI: [10.1016/0022-247X\(89\)90081-4](https://doi.org/10.1016/0022-247X(89)90081-4).
- [3] N. MacDonald, *Time Lags in Biological Models*, Lecture Notes in Biomathematics 27, Springer, 1978. DOI: [10.1007/978-3-642-93107-9](https://doi.org/10.1007/978-3-642-93107-9).